Workshop: Logical And Physical Architecture

<u>Case Study</u>:

<u>Background and Context</u>

After the Covid19 outbreak travelling has resumed and the global travel industry has been flourishing. The demand for the travel industry has changed drastically. There has been an increase of demands for **personalization, efficiency, and 24/7 availability**. Traditional travel agencies and online booking platforms often struggle to meet these needs because:

- They rely on manual agents with limited availability.
- Search interfaces overwhelm customers with options instead of guidance.
- Dynamic changes (flight delays, weather, cancellations) are poorly handled in real time.

Go East Holiday Inc, company that specializes holiday packages in east Asia countries such as **China, Japan, Korea, Hong Kong, Taiwan and Vietnam** recently secured a grant from a government agency to implement an **Agentic AI** travel system.

An **Agentic AI travel system** is designed to act as an **autonomous travel concierge**, offering planning, booking, and real-time adaptation, while collaborating with both humans and other digital agents (airlines, hotels, insurance providers).

Go East Holiday Inc awarded to a local AI company **Smart Michael System (SMS).** As an architect of **SMS**, you are about to begin your journey to design and implement this system.

Objectives

- **Personalization**: Tailor itineraries based on user preferences, budgets, and constraints.
- **Automation**: Handle multi-step processes like comparing flights, booking hotels, and generating itineraries.
- **Resilience**: Adapt to disruptions (e.g., flight delays, weather changes) in real time.
- **Transparency**: Provide explainable recommendations, including trade-offs (e.g., cost vs. convenience).
- **Integration**: Connect with external systems (Amadeus API, hotel booking engines, insurance providers, weather services).

Example User Journey

Scenario: **Amy**, a 32-year-old professional, wants a **5-day trip to Japan** next month.

1. **Conversation**:
   Amy: "I want a winter trip to Japan in December, under $2000, preferably including Kyoto and Tokyo."
2. **Planning**:
   - **The agent** should suggest:
     - Flight to Tokyo → 3 nights Tokyo cultural highlights → Bullet train to Kyoto → 2 nights Kyoto temples → Return flight.
   - Trade-offs explained: "Option A: cheaper flights with longer layovers; Option B: direct flight slightly above budget."
3. **Booking**:
   - **The agent** finalizes flights and hotels, applying any **Amy's** loyalty points automatically.
4. **Advisory Support**:
   - **The agent** provides visa guidance, Japan Rail Pass options, and recommended cultural etiquette.
5. **Real-Time Disruption**:
   - On travel day, a flight delay notification triggers the **Rescue Agent**, which proposes rerouting via Osaka. **Amy** approves via mobile.
6. **Post-Trip**:
   - The system logs feedback, stores preferences, and refines future recommendations.

Benefits

- **For Travelers**: Time savings, personalized trips, stress reduction.
- **For Agencies**: Cost reduction (fewer human agents needed) increased upsell opportunities, better customer retention.
- **For Partners**: Hotels, airlines, and insurers gain more targeted customer matches.

Non-Functional Requirements

- **Ethical**: Avoid bias in recommendations (e.g., always favouring certain airlines).
- **Privacy**: Secure storage of user data and compliance with GDPR/PDPA.
- **Trust**: Ensure explainability and transparency in why certain options are suggested.
- **Scalability**: Handle peak seasons with millions of concurrent users.
- **Resilience**: Ensure degraded-mode operations if APIs fail (e.g., suggest offline itineraries).

Workshop : Logical Architecture

The logical architecture should include the following

- Architecture Overview
    - Identify the Agents in the system and clearly describes their roles.
    - Identify the software components in the system and clearly describes their functions
    - The architecture style adopted
- Logical Component interaction Architecture
    - How are the components interacting with one another?

Workshop : Physical Architecture

The physical architecture should include the following

- Physical Architecture Overview
    - Identify the tech stack for each component or group of components.
    - Provide justification for some of the tech stack chosen and the physical design you have adopted.
- Cloud Architecture
    - Describe and illustrate clearly how the components (including cloud native services) are going to be hosted in the cloud.
    - Clearly justify the design and decision made in using certain components or tools.
    - The cloud architecture must be an end to end (from front end to back end)
    - Be sure to address all the non-functional requirements.